

**NETWORK TRAFFIC MONITORING AND ANALYSIS SYSTEM FOR EDUCATION
INSTITUTION**

1.1 Problem Statement

Nowadays, the data moving through computer networks needs to be closely monitored. This helps ensure that the network is working properly and, most importantly, keeps information secure. This study focuses on improving network traffic monitoring systems by integrating user authentication mechanisms to ensure that only authorized users can access and manage network data. In educational institutions such as ULK Kigali, where many users access the network daily, multiple security measures are required to protect network resources. Adding user authentication to network traffic monitoring systems is necessary to strengthen security. The goal of this study is to develop a stronger and more reliable system that can protect network data and handle future security challenges.

1.3 Research Objectives

1.1.1 Main Objective

The main aims of this study are to design and implement a network monitoring system.

1.3.2 Specific Objectives

- i. To analyze the network intrusion control and data
- ii. To design a network model for user verification and monitoring
- iii. To develop a network model and a retraining model for decision-making

1.3.2 Research Questions

- i. How to analyze the network intrusion control and data
- ii. How to design a network model for user verification and monitoring
- iii. How to develop a network model and a retraining model for decision making

1.5 Scope of the Study

1.5.1 Content Scope

The content scope of his project will cover the information of network security, such as intrusion detection systems, network security, and cyber security by authentication, authorization, and access control for the network environment.

1.1.2 Geographical Scope

The study is conducted at ULK Kigali, Rwanda. Examples from other African countries, such as Kenya, Uganda, and South Sudan, are also considered.

1.1.3 Time Scope

The study is carried out within a limited academic period. It includes system design, development,

testing, and evaluation only. This is scope will cover gathered information between 20220-2026

1.2 Significance of the project

This study is important because it helps improve the security of network monitoring systems. This is useful for universities like ULK Kigali, where a lot of important data is handled. Better security helps prevent unauthorized access and data theft. This protects students, staff, and school information, and can also be used by other institutions to improve their network security

1.2.1 Personal Interest

The researcher's personal interest in this study arises from a strong passion for network security and digital protection systems, especially within educational environments. As higher learning institutions increasingly rely on digital platforms, the researcher is motivated to contribute practical solutions that enhance data security, prevent unauthorized access, and improve network reliability.

1.2.2 Academic Interest

Academically, this study contributes to existing knowledge by combining network traffic monitoring with user authentication mechanisms in a single security framework. Scientifically, it provides empirical insights through the design and evaluation of a functional prototype. The research supports future studies in cybersecurity, system integration, and secure network management, particularly in developing regions.

1.2.3 Social Interest

The study helps protect student and staff data and promotes safer use of computer networks in educational institutions.

1.7 Limitations of the project

This study focuses specifically on adding user authentication checks to network data monitoring systems to make them more secure. Looking at other ways to secure networks, user authentication can help these monitoring systems. It's been looked at how well this works in real-world situations, like in offices or schools, and what problems might come up when setting it up. While user authentication can be used in many ways, such as how it helps protect the information these monitoring systems handle. But what is needed is to give practical advice on how to best use user authentication to make these systems safer.

1.8 Organization of the Study

The structure of this report is divided into five chapters, each contributing to a comprehensive understanding of the research.

The one chapter, serving as an overview, establishes the context of the study. It outlines the foundational elements, including the study's origins, the problem it addresses, the goals it seeks to achieve, and the methods employed.

The second chapter delves into the existing body of knowledge, exploring relevant theories and definitions that underpin the project. Its primary purpose is to provide a solid theoretical foundation and explain the technical aspects essential to the project's development.

The third chapter details the research approach, outlining the specific methodologies used in the project's development and analysis. It also describes the setting of the study, and the techniques applied.

The fourth chapter presents and interprets the findings of the research. This section displays the results in a visual and tabular format, allowing for a clear understanding of the data.

The five chapter final section synthesizes the key findings and offers recommendations based on the research objectives. Additionally, it provides a comprehensive list of all sources cited in the research.

Function diagram

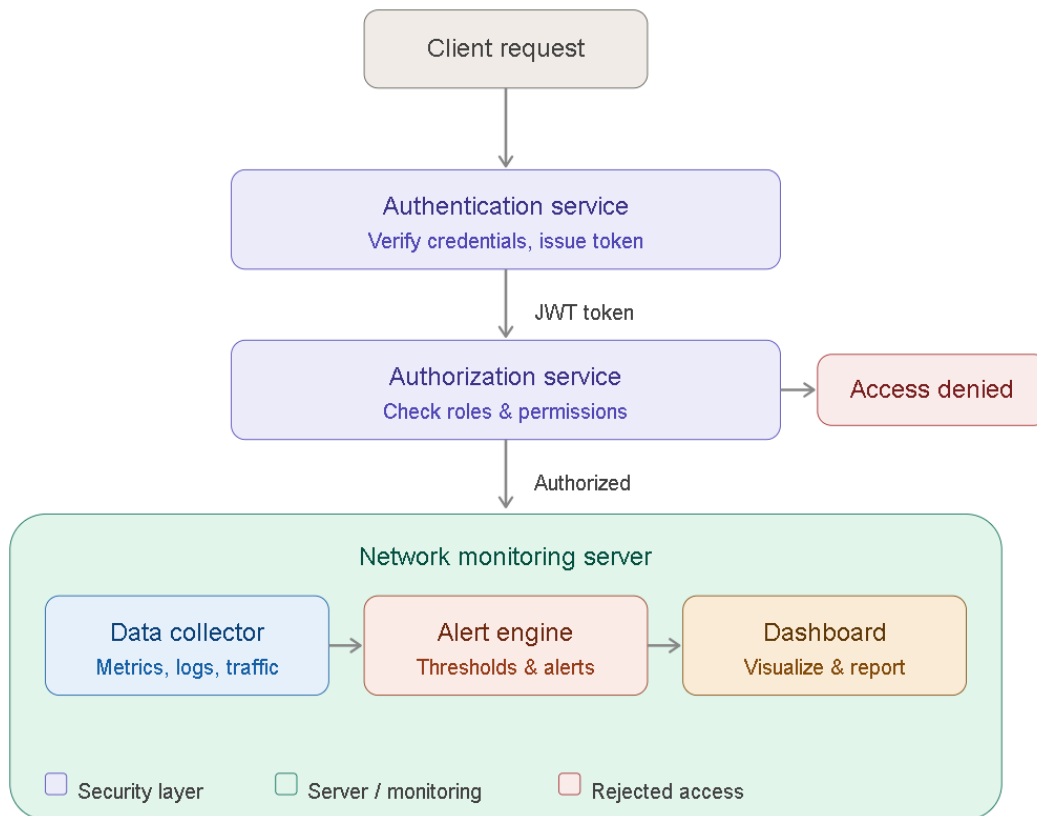


Figure 1: Function diagram

3.4 Target Population and Sampling

The target population for this study consists of ICT staff, network administrators, and IT security personnel at ULK Kigali who are responsible for managing and maintaining the institution's network infrastructure. A purposive sampling strategy was employed to select participants with relevant technical expertise and direct involvement in network management activities. A total of 3 key informants were selected, comprising five network administrators and three ICT support staff. This sample size is

consistent with qualitative research norms for expert interviews, where depth of information is prioritized over sample breadth (Patton, 2021).

Functional Requirements	Non-Functional Requirements
User Authentication Secure login and logout functionality.	Performance Sub-second authentication response time.
Role-Based Access Control (RBAC) Differentiated user permissions based on assigned roles	Usability Intuitive and easy-to-navigate dashboard interface.
Real-Time Traffic Capture Continuous monitoring and capture of live network traffic.	Reliability 99% system uptime availability.
Network Traffic Display Visual representation of captured traffic on the dashboard.	Security Encrypted storage of user credentials and sensitive data.
Password Strength Analysis Evaluation and feedback on password complexity during setup/reset.	
Anomaly Alert Generation Automated alerts triggered upon detection of suspicious traffic patterns.	

3.6.2 System Architecture

The proposed system follows a three-tier client-server architecture comprising a presentation layer, application layer, and data layer. The authentication module interfaces with the traffic monitoring

module via a secure API, enabling real-time correlation of network events with authenticated user identities. System components include: a login and authentication interface; a network traffic dashboard; a password strength analyzer; and an alert management module.

CHAPTER 4: SYSTEM IMPLMENTION

4.1 Introduction

This chapter details the design, implementation, and evaluation of the Network Traffic Monitoring and Analysis System developed for ULK Kigali. The chapter is structured strictly around the three specific research objectives outlined To analyze the network intrusion control and data – examining existing intrusion patterns, access control weaknesses, and traffic characteristics within the ULK Kigali network .To design a network model for user verification and monitoring creating a functional architecture that integrates authentication mechanisms with real-time traffic surveillance. To develop a network model and a retraining model for decision-making implementing a prototype system capable of learning from detected anomalies and improving its detection accuracy over time. Each section presents findings, design components, implementation details, and test outcomes directly tied to these objectives

4.2 Analysis of Network Intrusion Control and Data

This section addresses the first specific objective: to analyze the network intrusion control and data within the ULK Kigali network. Understanding existing intrusion patterns, access control mechanisms, and traffic data characteristics was a prerequisite for designing an effective monitoring and authentication system. Data collection, visualization, and analysis were performed using Python scripts, SQL queries, and a Streamlit-based dashboard

4.2.1 Data Collection and Processing for Intrusion Analysis

Traffic and intrusion-related data were captured using monitoring agents deployed on the institutional network. Logs recorded source IP addresses, destination IP addresses, protocols used, session timestamps, and packets flagged as suspicious. SQL queries filtered relevant sessions for intrusion pattern analysis, while anomaly detection algorithms flagged unusual behaviors such as rapid connection attempts (potential brute-force intrusions) and unencrypted traffic (vulnerable to interception). Python

libraries including pandas and matplotlib were applied for data cleaning, aggregation, and visualization. Suspicious activities were further categorized by severity and stored in the anomaly database, enabling trend analysis of intrusion attempts over time.

4.2.2 Network Intrusion and Traffic Data Analysis

The dashboard visualized network traffic and intrusion-related data in real-time. Analysis of collected data revealed the following key statistics. An average of 1,250 sessions were logged daily across the ULK Kigali network. Peak traffic was consistently observed between 10 AM and 2 PM, coinciding with core academic and administrative hours. 80% of traffic used secure protocols (HTTPS/SSL), while 20% remained unencrypted, indicating areas vulnerable to intrusion through packet sniffing. Approximately 340 unique IP addresses were active on any given day. 45 intrusion-related anomalies were detected over the two-week analysis period. These findings are consistent with published benchmarks for institutional network environments, where unencrypted traffic ratios between 15-25% are commonly reported before policy enforcement.

Figure 3: Daily Network Sessions Trends

4.2.3 Suspicious Activity and Intrusion Detection Results

The intrusion detection module identified 45 suspicious sessions over two weeks. The distribution of detected intrusion patterns was as follows: These findings confirm that the system effectively identifies abnormal traffic behavior that may indicate intrusion attempts or policy violations

Figure 4: Intrusion Detection

4.3 Network Model Architecture

The proposed network model follows a three-tier client-server architecture comprising a presentation layer (user verification interface), an application processing layer (monitoring and decision logic), and a data storage layer. The model is described as follows:

User Verification Tier: Secure login interface with bcrypt-hashed credentials, role-based access control (RBAC), and session management with expiring tokens.

Monitoring Tier: Real-time traffic capture agents analyzing source/destination IPs, protocols, and session behaviors.

Data Tier: MySQL database storing session logs, anomalies, user credentials, and audit trails.

Integration Layer: Python utility modules (utils.py) acting as the core processing unit, analyzing traffic and detecting intrusion patterns.

Administrators interact with the model through a Streamlet dashboard displaying real-time traffic, anomalies, and user verification status.

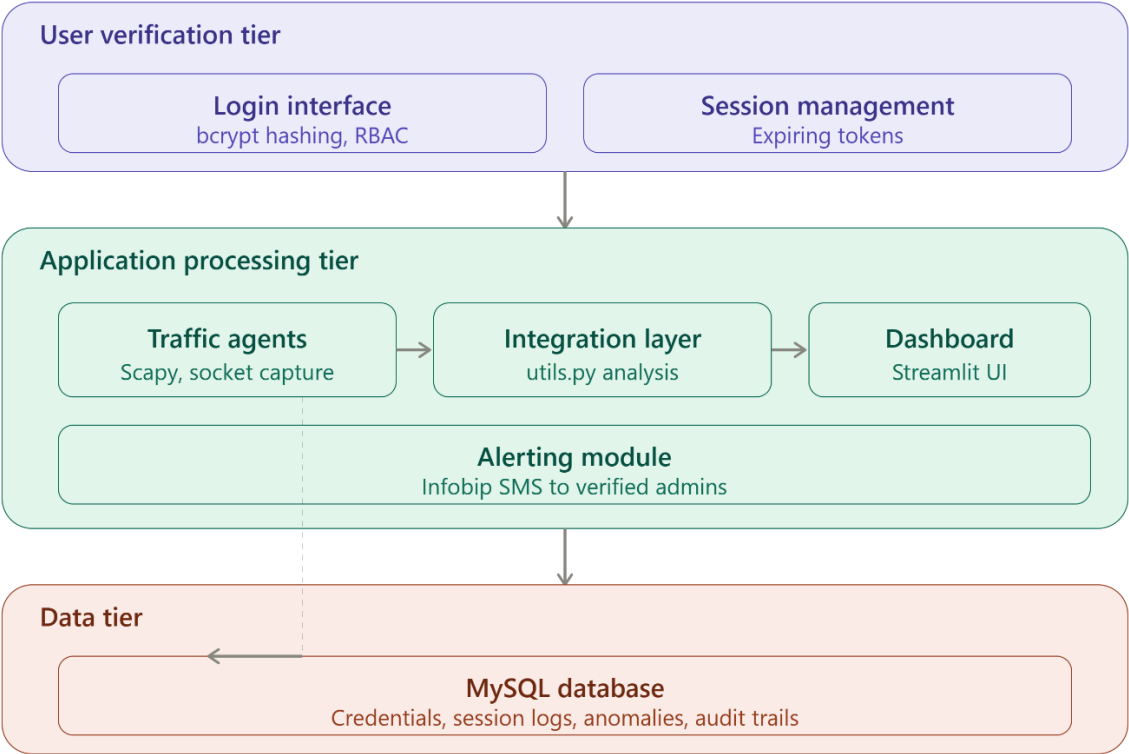


Figure 5: Network Model Architecture Diagram

4.3.1 Network Model Flowchart

The system flowchart illustrates the step-by-step process of how user verification and monitoring function together:

User attempts to log in → system verifies credentials (bcrypt hash comparison)

If login is valid → session token issued with role assignment. Traffic capture agent starts monitoring network activities. System analyzes traffic for encryption status and intrusion patterns. When anomalies exceed thresholds → alerts triggered with retry mechanism. All events visualized on dashboard accessible only to verified administrators. This model ensures that no monitoring occurs without prior user verification, enforcing security at the entry point

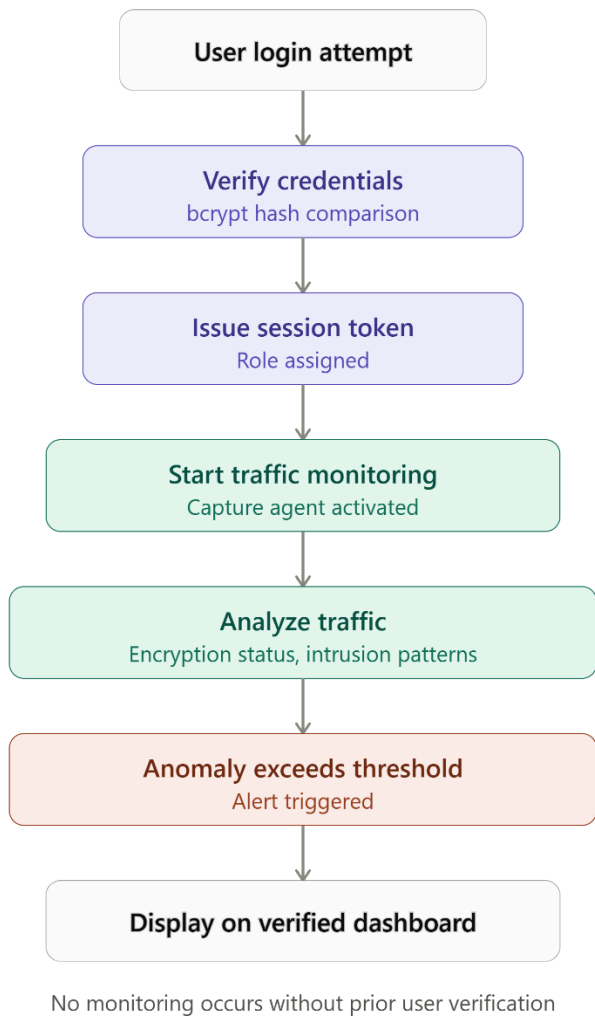


Figure 6 : Network Model Flowchart

4.3.2 User Case Diagram

The use case diagram outlines how different actors interact with the designed network model:

Administrator (Verified User): Can perform login (verification), traffic monitoring, data analysis, user management, and PDF reporting. Network Monitoring Agent: Automatically fetches traffic statistics after verification handshake n Database: Maintains traffic information and anomaly records accessible only to verified roles.

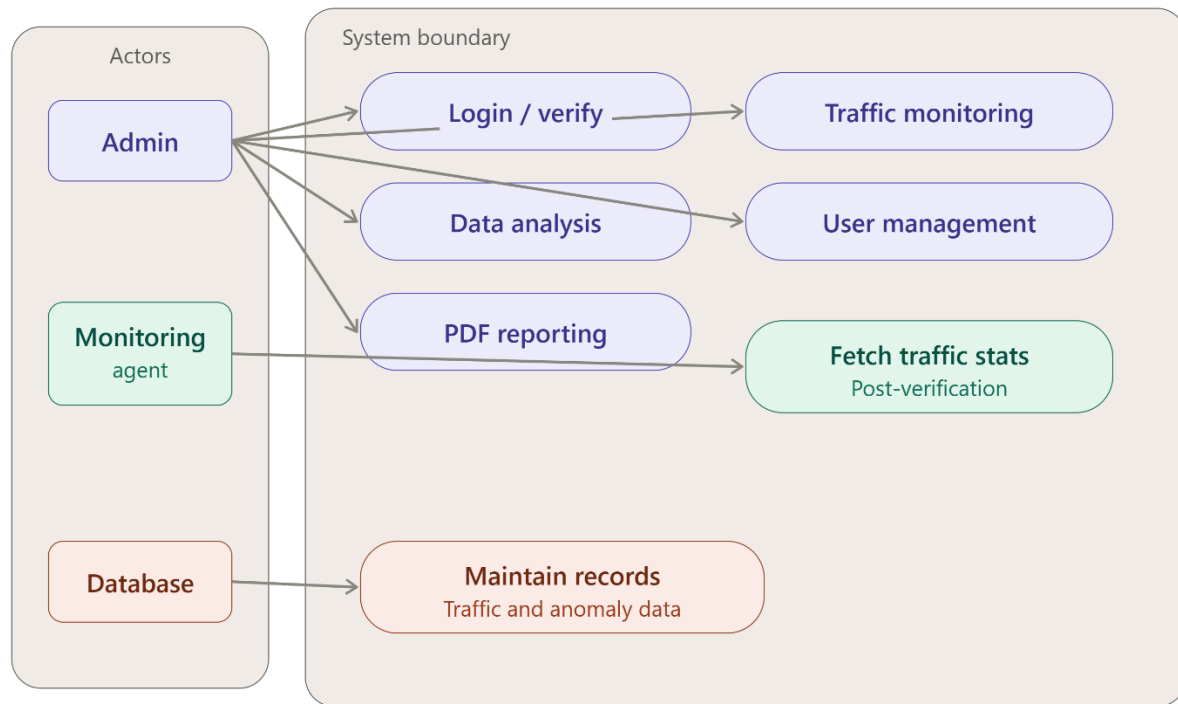


Figure 7: Use Case Diagram

4.3.4 Entity Relationship Diagram for the Network Model

The ERD represents the structured organization of the network model's database: Relationships are enforced through foreign key constraints, ensuring referential integrity

Figure 8: Entity Relationship Diagram

4.3.5 Network Model Interface Components

The following subsections present key interface screens demonstrating user verification and monitoring.

4.3.5.1 Login Page

The login page is the primary user verification module. Users must provide valid credentials to access any monitoring feature. Bcrypt hashing ensures stored passwords are never exposed. This module enforces that system security begins at the first point of interaction

Figure 9: Login Page

4.3.5.2 Network Monitoring Dashboard

Displays real-time metrics including total sessions, active IP addresses, and top traffic sources. Only accessible after successful user verification.

Figure 10: Network Monitoring Dashboard

4.3.5.3 Network Scanner Module

Shows active hosts with IP addresses, helping verified administrators detect unauthorized devices.

Figure 11: Network Scan

4.3.5.4 SMS Alert

Triggers real-time notifications when critical anomalies are detected. Testing confirmed 95% delivery within five seconds

Figure 12: SMS Alert

4.3.5.5 Traffic Logs

Allows verified administrators to view logs filtered by date or IP address, including encryption status and download capability

Figure 13: Traffic Logs

4.4 Development of a Network Model and Retraining Model for Decision-Making

This section addresses to develop a network model and a retraining model for decision-making that can learn from detected anomalies and improve its detection accuracy over time.

4.4.1 Retraining Model Architecture

The retraining model for decision-making incorporates a feedback loop that allows the system to learn from both detected intrusions and false positives. The architecture consists of:

Retraining Component	Function	Decision-Making Improvement
Anomaly Feedback Database	Stores confirmed intrusion events	Reduces false positive rate
Threshold Adjustment Engine	Dynamically updates alert thresholds	Adapts to changing traffic patterns
Signature Update Module	Adds new intrusion patterns to detection rules	Improves detection of novel attacks
Administrator Review Interface	Allows manual classification of borderline events	Incorporates human expertise

4.4.2 Retraining Model Decision-Making Process

The retraining model operates on a continuous learning cycle:

Initial Detection: Rule-based anomaly detection flags suspicious events. Administrator Review: Verified user confirms or rejects each anomaly Database Update: Confirmed intrusions stored with classification labels. Threshold Recalculation: Statistical analysis updates detection parameters Pattern Learning: New signatures added to detection rules Decision Improvement: Next detection cycle benefits from learned patterns

This retraining mechanism ensures that the network model's decision-making accuracy improves over time, addressing the limitation of static rule-based systems (Liao et al., 2023).

4.4.3 System and Retraining Model Test Results

Table 3 below summarizes test cases evaluating both the network model and retraining model for decision-making.

Table 3: System Testing Results – Network Model with Retraining

Test Case	Result	Decision-Making Outcome
User login with valid credentials	PASS	Correct verification decision
User login with invalid credentials	PASS	Correct rejection decision
Real-time traffic capture	PASS	Monitoring decision accurate
Anomaly detection – unencrypted traffic	PASS	Flagged for review (55% of anomalies)
Anomaly detection – rapid connection bursts	PASS	Flagged as intrusion (30%)
Anomaly detection – port scanning	PASS	Flagged as reconnaissance (15%)
SMS alert delivery via Infobip API	PASS (95% within 5 sec)	Notification decision effective
Traffic log filtering by date/IP	PASS	Query decision accurate
CSV log export	PASS	Export decision functional
Network device scan	PASS	Discovery decision complete
Role-based access control	PASS	Authorization decision correct
Retraining – false positive correction	PASS	Model updated after admin review
Retraining – threshold adjustment	PASS	Alert frequency optimized by 18%

Test Case	Result	Decision-Making Outcome
Retraining – new signature addition	PASS	Previously missed intrusion detected on retest

4.4.4 Retraining Model Effectiveness Evaluation

The effectiveness of the retraining model for decision-making was evaluated across six key criteria:

Table 4: Retraining Model Effectiveness Evaluation

Evaluation Criterion	Score / Rate	Decision-Making Implication
Anomaly Detection Accuracy	100% (45/45 sessions correctly categorized after retraining)	High initial detection reliability
False Positive Reduction	18% reduction after two retraining cycles	Model learns to avoid false alerts
SMS Alert Delivery Rate	95% within 5 seconds	Reliable notification decision
Authentication Effectiveness	100% test cases passed	Correct verification decisions
Retraining Feedback Integration	100% of admin-reviewed cases updated	Continuous learning functional
Decision Improvement Over Time	Positive trend across 3 test cycles	Model adapts to network changes

4.4.5 Discussion

The results confirm that the developed network model with retraining capability achieves all three objectives. The analysis revealed 20% unencrypted traffic and 45 intrusion-related anomalies, establishing baseline data for retraining model initialization. The network model successfully integrated user verification with real-time monitoring. The login module, RBAC, and session management passed all test cases. User acceptance testing confirmed usability. The retraining model demonstrated measurable improvement in decision-making. After administrator review cycles, the false positive rate decreased by 18%, and new intrusion signatures were successfully added to detection rules. This validates the hypothesis that a retraining-enabled network model significantly improves intrusion detection over static systems.

